TESTLEAF TRAINING

BASIC JAVASCRIPT PROGRAMS

Interview Preparation – Practice Questions

Questions

1. Write a JavaScript program to check whether a number is even or odd.

2. Write a JavaScript program to check whether a number is positive, negative, or zero.

3. Write a JavaScript program to find the largest of three numbers.

4. Write a JavaScript program to check whether a year is a leap year.

5. Write a JavaScript program to calculate the factorial of a number.

6. Write a JavaScript program to print the Fibonacci series up to a given number of terms.

7. Write a JavaScript program to check whether a string is a palindrome.

8. Write a JavaScript program to reverse a string without using the built-in reverse() method.

9. Write a JavaScript program to check whether a number is a prime number.

10. Write a JavaScript program to find the sum of the digits of a number.

Answer 1: Even or Odd

Program

```
const number = 17;

if (number % 2 === 0) {
    console.log(`${number} is even`);
} else {
    console.log(`${number} is odd`);
}
```

Expected Output: 17 is odd

Explanation: The modulus operator (%) returns the remainder. If the remainder after division by 2 is 0, the number is even.

Answer 2: Positive, Negative, or Zero

Program

```
const number = -5;

if (number > 0) {
    console.log("The number is positive");
} else if (number < 0) {
    console.log("The number is negative");
} else {
    console.log("The number is zero");
}
```

Expected Output: The number is negative

Explanation: The program checks whether the value is greater than zero, less than zero, or neither.

Answer 3: Largest of Three Numbers

Program

```
const a = 12;
const b = 27;
const c = 19;

if (a >= b && a >= c) {
    console.log(`${a} is the largest`);
} else if (b >= a && b >= c) {
    console.log(`${b} is the largest`);
} else {
    console.log(`${c} is the largest`);
}
```

Expected Output: 27 is the largest

Explanation: Each condition compares one value with the other two. Using >= also handles cases where values are equal.

Answer 4: Leap Year

Program

```
const year = 2024;

if ((year % 400 === 0) || (year % 4 === 0 && year % 100 !== 0)) {
    console.log(`${year} is a leap year`);
} else {
    console.log(`${year} is not a leap year`);
}
```

Expected Output: 2024 is a leap year

Explanation: A year is a leap year if it is divisible by 400, or divisible by 4 but not by 100.

Answer 5: Factorial of a Number

Program

```
const number = 5;
let factorial = 1;

if (number < 0) {
    console.log("Factorial is not defined for negative integers");
} else {
    for (let i = 1; i <= number; i++) {
        factorial *= i;
    }
    console.log(`Factorial of ${number} is ${factorial}`);
}
```

Expected Output: Factorial of 5 is 120

Explanation: Factorial is the product of all positive integers from 1 to the given number. By convention, 0! is 1.

Answer 6: Fibonacci Series

Program

```
const terms = 7;
let first = 0;
let second = 1;
const series = [];

for (let i = 0; i < terms; i++) {
    series.push(first);
    const next = first + second;
    first = second;
    second = next;
}

console.log(series.join(" "));
```

Expected Output: 0 1 1 2 3 5 8

Explanation: Each term after the first two is the sum of the two previous terms. This example prints seven terms.

Answer 7: Palindrome String

Program

```
const text = "madam";
const reversed = text.split("").reverse().join("");

if (text === reversed) {
    console.log(`${text} is a palindrome`);
} else {
    console.log(`${text} is not a palindrome`);
}
```

Expected Output: madam is a palindrome

Explanation: A palindrome reads the same forward and backward. This solution uses reverse() for clarity; see Question 8 for reversing without it.

Answer 8: Reverse a String Without reverse()

Program

```
const text = "JavaScript";
let reversed = "";

for (let i = text.length - 1; i >= 0; i--) {
    reversed += text[i];
}

console.log(reversed);
```

Expected Output: tpircSavaJ

Explanation: Start from the last character and append each character to a new string while moving backward.

Answer 9: Prime Number

Program

```
const number = 29;
let isPrime = number > 1;

for (let i = 2; i <= Math.sqrt(number) && isPrime; i++) {
    if (number % i === 0) {
        isPrime = false;
    }
}

console.log(isPrime ? `${number} is a prime number` : `${number} is not a prime number`);
```

Expected Output: 29 is a prime number

Explanation: A prime number is greater than 1 and has no divisors other than 1 and itself. Checking divisors up to the square root is sufficient.

Answer 10: Sum of Digits

Program

```
let number = 4825;
let sum = 0;
number = Math.abs(number);

while (number > 0) {
    sum += number % 10;
    number = Math.floor(number / 10);
}

console.log(`Sum of digits is ${sum}`);
```

Expected Output: Sum of digits is 19

Explanation: The modulus operator extracts the last digit; Math.floor(number / 10) removes it. Repeat until no digits remain.